

Report: Optimising NYC Taxi Operations

Include your visualisations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

1.1.1. Sample the data and combine the files

To obtain a representative yet computationally efficient subset of trip records while preserving the natural temporal distribution of taxi activity, a **stratified random sampling methodology** was applied. The complete process is summarised below.

The data is organised as **monthly parquet files** (e.g., 2023-01.parquet), each containing detailed Yellow Taxi trip records

- 1) Read Monthly Files:
Each month's parquet file is loaded by looping over files. From the pickup time, we extract from "tpep_pickup_datetime":
 - the **date**
 - the **hour of the day**
- 2) Iterate by date, then by hour:
 - Ensures sampling is performed for each date separately, preserving the daily structure
 - For each date, loops through all 24 hours. So that dataset was filtered to include only the records belonging to each specific hour.
- 3) Hourly sampling:
 - Within each hour, 0.008% of the available records were randomly selected using a randomised selection method (`sample(frac=0.008, random_state=42)`) to ensure consistency and reproducibility of results.
 - `Frac = 0.008` is intentional, resulting in approximately 300,000 records.
 - Sampling by hour preserves the temporal distribution (time-of-day effects, rush hours, overnight patterns) so the sample is representative across hours and dates, not biased toward heavily recorded hours.
- 4) Data Aggregation:
After processing all dates/hours in the month, the month's hourly samples are concatenated into the global dataframe df.

Strengths of this approach:

- Temporal representativeness: by sampling each hour separately, you keep the diurnal patterns (rush hours, night behaviour).
- I also ran a code to check if the sample is perfectly random, where I managed to get 7-9% data from every parquet file.
- Reproducibility: fixed fraction and random seed give identical samples on re-run.

2. Data Cleaning

2.1. Fixing Columns

2.1.1. Fix the index

Below, unnecessary columns were dropped:

- 'Unnamed: 0': Unnamed column with index numbering, find it useless.
- `store_and_fwd_flag`: Not useful for analysis; indicates if the trip record was held in vehicle memory before sending to the server.
- `mta_tax`: Fixed tax amount, does not provide variability.
- `tolls_amount`: Not relevant for most trips, as tolls are not always applicable.
- `date`: column created during the process of sampling.

2.1.2. Combine the two airport_fee columns

- There are two airport fee columns: `airport_fee` and `Airport_fee`. This may be an error in the naming of the columns..
- The `airport_fee` column has more null values than the `Airport_fee` column, so we can remove the `airport_fee` column and keep the `Airport_fee` column.
- To combine these columns, we used the `fillna()` method from pandas, which is designed to replace NaN values. In this case, it uses the values from the `Airport_fee` column to fill the NaN entries in the `airport_fee` column..

2.1.3. Fix columns with negative (monetary) values

Identify financial columns such as `['extra', 'improvement_surcharge', 'total_amount', 'congestion_surcharge', 'airport_fee']` that have unexpected negative values in the dataset. As there are only a few negative records, remove all negative values while keeping NaN values intact.

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column

To ensure data integrity, missing values in critical columns were identified and addressed. All of these columns have 3.35% missing values.

2.2.2. Handling missing values in passenger_count

Fill all missing values and unwanted zero passenger count values with a median of 1.0, as zero passenger count rows indicate some fare amount, suggesting there must be passengers.

2.2.3. Handle missing values in RatecodeID

I replaced all missing values with the mode, as this column is categorical..

2.2.4. Impute NaN in congestion_surcharge.

All missing values were replaced with 0; the constant values found were 2.5 and 0.

airport_fee : All missing values for airport fees have been replaced with 0, assuming that there were no visits to the airport.

2.3. Handling Outliers and Standardising Values

2.3.1. Check outliers in payment type, trip distance and tip amount columns

To improve data quality and ensure meaningful analysis, outliers were identified and removed based on domain knowledge and logical constraints. The following steps were taken:

- 1) **passenger_count**: Entries where the passenger count was greater than 6 or equal to 0 were removed to ensure realistic trip records.
 - 2) Entries where **trip_distance** is nearly 0 and **fare_amount** is more than 300
 - 3) Entries where trip_distance and fare_amount are 0 but the pickup and dropoff zones are different (both distance and fare should not be zero for different zones)
 - 4) Entries where trip_distance is more than 250 miles.
 - 5) Entries where **payment_type** is 0 (there is no payment_type= 0 defined in the data dictionary)
 - 6) In the "fare_amount" and "total_amount" columns, we found extreme records that we have removed.
 - 7) Calculating **trip_duration** and removing negative values.
- No further standardisation was required because the analysis relies primarily on correlation, aggregation, and visualization techniques, which are not affected by differences in variable scale. Additionally, outliers were already treated, and keeping values in their original units (miles, dollars, hours) preserves business interpretability.
 - Standardisation would only be necessary if distance-based machine learning models were applied.

3. Exploratory Data Analysis

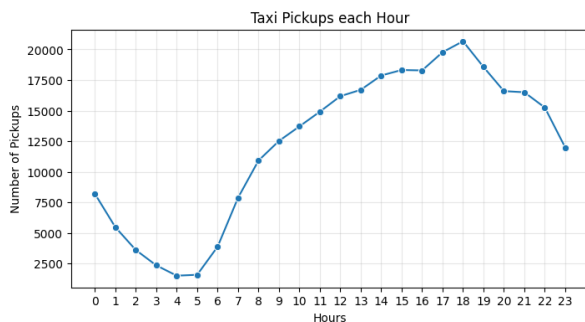
3.1. General EDA: Finding Patterns and Trends

3.1.1. Classify variables into categorical and numerical

Categorical	Numerical
VendorID	fare_amount
tpep_pickup_datetime	extra
tpep_dropoff_datetime	mta_tax
passenger_count	tip_amount
trip_distance	tolls_amount
RatecodeID	improvement_surcharge
PULocationID	total_amount
DOLocationID	congestion_surcharge
payment_type	airport_fee
pickup_hour	
trip_duration	

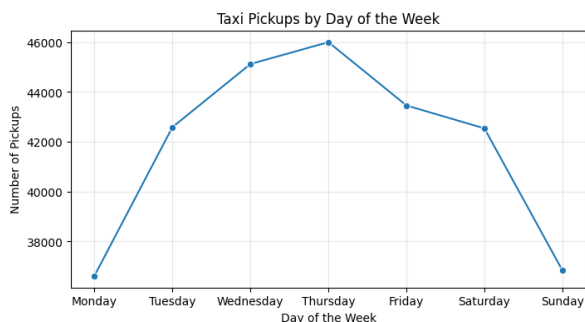
- A new column "trip_duration" will be created in minutes, based on the tpep_pickup_datetime and tpep_dropoff_datetime columns. We will remove rows with negative trip durations and extreme outliers.

3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months



Hourly Pickups:

The busiest hours for taxi pickups are consistently in the late afternoon and evening (between 5:00 PM and 7:00 PM), peaking around 6 PM (18:00) with over 20,000 trips in the sampled data, scaling up to approximately 2.5 million actual trips.



Daily Pickups:

Mid-week days (Tuesday, Wednesday, Thursday, Friday) exhibit the highest number of pickups, with Thursday having the highest count (46,004 trips).

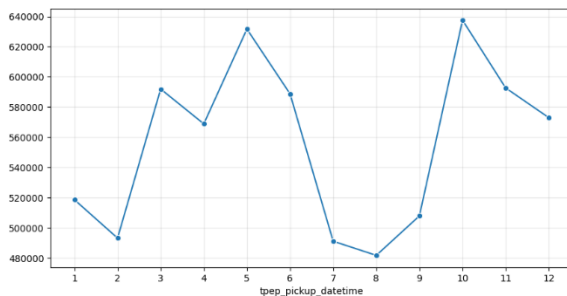


- Monthly Pickups:** There isn't a dramatic variation in monthly pickups, but **May (27,063 trips)** and **October (26,692 trips)** generally see higher activity, while July, August, and September show slightly lower counts. January and February also have lower pickup counts.

3.1.3. Filter out the zero/negative values in fares, distance and tips

Created a new dataframe by removing all negative values from financial parameters, such as fare amount, tip amount, total amount, and trip distance.

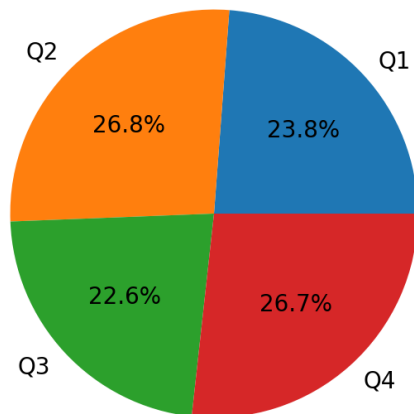
3.1.4. Analyse the monthly revenue trends



Monthly revenue closely follows the pickup trends, with **May and October** showing the highest revenue. The lowest revenue months are **February and August**. The total annual revenue is approximately \$6.67 million for the sampled This indicates seasonal variations in taxi demand and revenue.

3.1.5. Find the proportion of each quarter's revenue in the yearly revenue

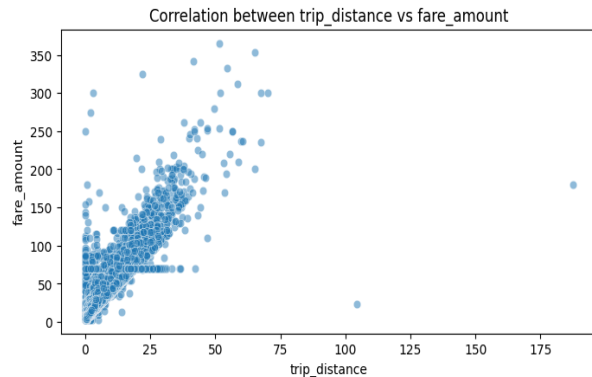
Proportion of Total Revenue by Quarter



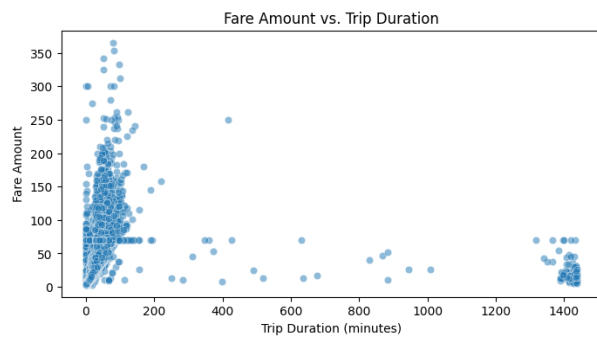
The revenue is fairly distributed across quarters, with Q1 (Jan-Mar) : 24.0%, Q2 (**Apr-Jun**) 26.8%, Q3 (Jul-Sep) 22.2%, Q4 (**Oct-Dec**) 27.0%

of the total annual revenue. Q2 and Q4 show slightly higher proportions.

3.1.6. Analyse and visualise the relationship between distance and fare amount

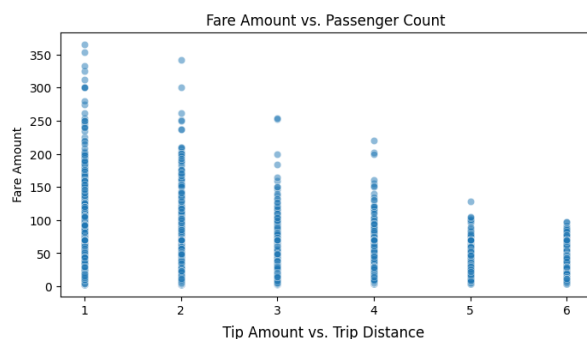


- There is a clear **positive correlation (0.94)**: longer trips generally have higher fares.
- Some fare variability exists for similar distances, likely due to traffic, tolls, or surcharges.
- Most trips fall within 0–20 miles and \$0–\$200, showing that typical taxi rides are short and moderately priced.
- A few outliers appear at very high distances or fares, representing unusual long trips or irregular fare values.

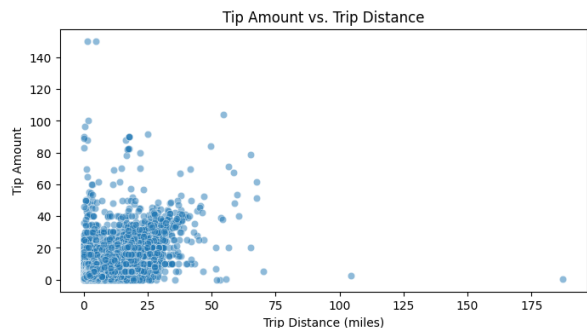


A **moderate positive correlation (0.29)** suggests that longer trip durations, even for similar distances, can lead to higher fares, likely due to waiting times or slower traffic.

3.1.7. Analyse the relationship between fare/ passengers and Tip/Distance

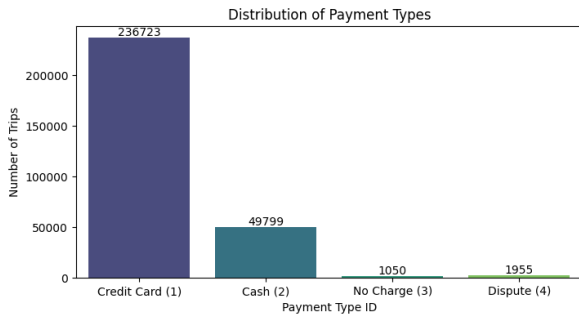


A **very weak positive correlation (0.04)** suggests that the number of passengers has minimal direct impact on the `fare\_amount` in general.



A **moderate positive correlation (0.59)** indicates that longer trips tend to result in higher tip amounts, which could be due to higher total fares or perceived greater service value.

3.1.8. Analyse the distribution of different payment types



Credit card payments (Type 1) are overwhelmingly the most common (226195 trips), All other Payment Types like cash (Type 2). 'Dispute' (Type 4) and 'No charge' (Type 3) are significantly less frequent.

3.1.9. Load the taxi zones shapefile and display it

I loaded all the geographical taxi zone shapefiles using the GeoPandas library.

3.1.10. Merge the zone data with trips data

Merging taxi trip data with taxi zone information using location IDs (**PULocationID** and **LocationID**). A left join keeps all rows from taxi trip data.

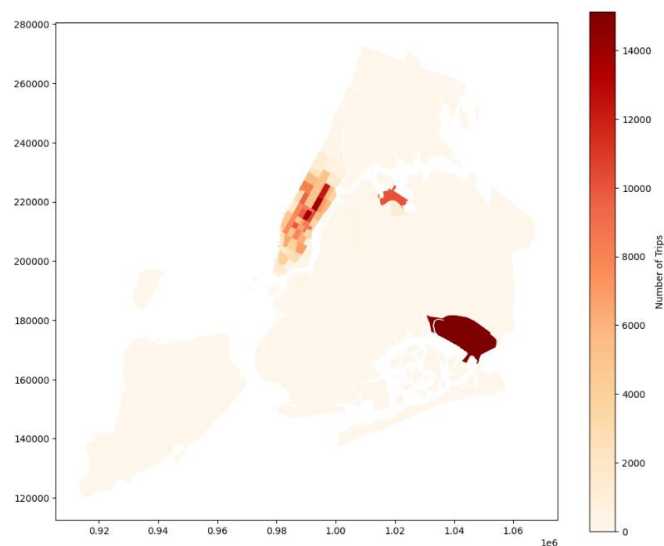
3.1.11. Find the number of trips for each zone/location ID

group taxi trips by their pickup location ID, counts the number of trips for each location, and then presents these counts in a new DataFrame

3.1.12. Add the number of trips for each zone to the zones dataframe

code merges taxi zone data (zones) with trip counts per location (trips_per_location), filling any missing trip counts with zero, and stores the result in zones_with_trips

3.1.13. Plot a map of the zones showing number of trips



3.1.14. Conclude with results

Busiest Zones (by Trip Count):

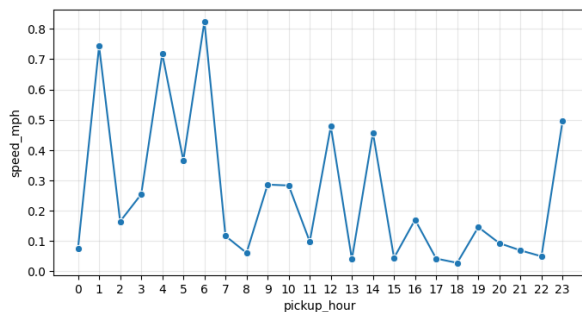
- **JFK Airport** is by far the busiest zone with **15,135 trips**.
- Following closely are Upper East Side South (13,788 trips), Midtown Center (13,625 trips), and Upper East Side North (12,329 trips). Midtown East also shows high activity with 10,573 trips.
- These zones represent major transportation hubs and dense urban areas, indicating high demand for taxi services.

Least Busy Zones (by Trip Count):

- Conversely, several zones recorded **zero trips**, indicating they are either less accessible, less populated, or not typical taxi pickup points in the sampled data.
- Zones like Stapleton, Broad Channel, Breezy Point/Fort Tilden/Riis Beach, West Brighton, Jamaica Bay, etc

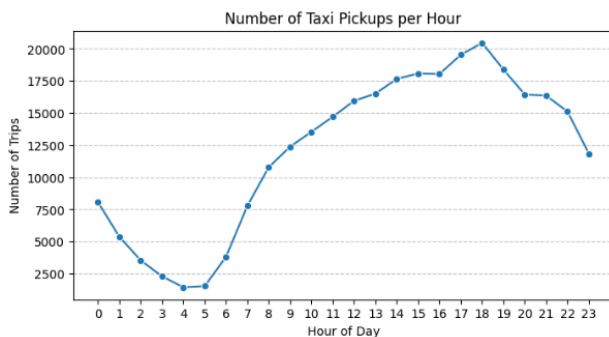
3.2. Detailed EDA: Insights and Strategies

3.2.1. Identify slow routes by comparing average speeds on different routes



- The speed remains relatively constant, fluctuating between approximately 0.036 and 0.825 miles per hour.
- The slowest routes do not vary significantly in speed throughout the day, indicating consistent traffic conditions on these routes.

3.2.2. Calculate the hourly number of trips and identify the busy hours



- The busiest hour is 6 PM (**18:00**), underscoring the evening rush as the period of highest demand.

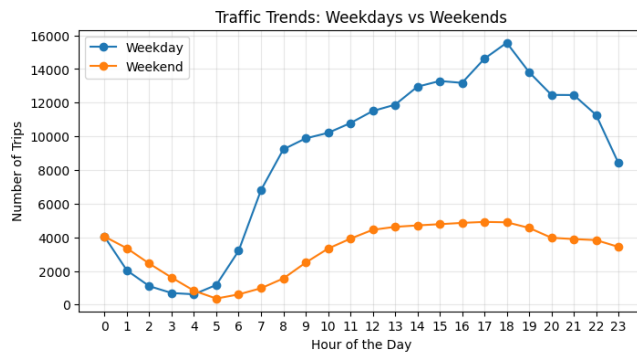
3.2.3. Scale up the number of trips from above to find the actual number of trips

Top 5 busiest hours:

pickup_hour	count
18:00	2,556,625
17:00	2,441,375
19:00	2,297,250
15:00	2,258,750
16:00	2,254,500

- Highlighting immense demand during evening times.

3.2.4. Compare hourly traffic on weekdays and weekends



Weekdays: Show two distinct peaks: a morning peak (around 8-10 AM) and a more pronounced evening peak (4-7 PM), corresponding to commuting patterns.

Weekends: Exhibit a flatter morning profile, with demand gradually rising throughout the day and peaking later in the evening (7-10 PM), reflecting leisure and social activities.

3.2.5. Identify the top 10 zones with high hourly pickups and drops

- Top 10 Pickup Zones:

	PULocationID	Zone	Trip_count
115	132	JFK Airport	15135
211	237	Upper East Side South	13788
143	161	Midtown Center	13625
210	236	Upper East Side North	12329
144	162	Midtown East	10573
121	138	LaGuardia Airport	10093
164	186	Penn Station/Madison Sq West	9989
204	230	Times Sq/Theatre District	9695
125	142	Lincoln Square East	9575
152	170	Murray Hill	8566

These are major hubs and dense urban areas.

- Top 10 Dropoff Zones:

	DOLocationID	zone	trip_count
8191	236	Upper East Side South	1957
8271	237	Upper East Side North	1719
8272	237	Upper East Side South	1331
8190	236	Upper East Side North	1287
5529	161	Upper East Side South	953
8245	237	Midtown Center	953
8170	236	Midtown Center	791
8392	239	Lincoln Square East	774
4804	142	Upper West Side South	758
8346	238	Upper West Side South	730

3.2.6. Find the ratio of pickups and dropoffs in each zone

Analysis of this ratio helps identify zones that are primarily origins (high ratio) or destinations (low ratio), which is critical for efficient cab positioning.

- Top 10 pickup dropoff ratio:

	LocationID	pickups	dropoffs	pickup_dropoff_ratio	zone	borough
69	70.0	1326.0	145	9.144828	East Elmhurst	Queens
125	132.0	15135.0	3040	4.978618	JFK Airport	Queens
131	138.0	10093.0	3426	2.946001	LaGuardia Airport	Queens
178	186.0	9989.0	6509	1.534644	Penn Station/Madison Sq West	Manhattan
41	43.0	4963.0	3612	1.374031	Central Park	Manhattan
239	249.0	6472.0	4760	1.359664	West Village	Manhattan
108	114.0	3748.0	2804	1.336662	Greenwich Village South	Manhattan
155	162.0	10573.0	8331	1.269115	Midtown East	Manhattan
154	161.0	13625.0	11419	1.193187	Midtown Center	Manhattan
98	100.0	4749.0	3997	1.188141	Garment District	Manhattan

- Bottom 10 pickup dropoff ratio:

	LocationID	pickups	dropoffs	pickup_dropoff_ratio	zone	borough
6	8.0	0.0	5	0.0	Astoria Park	Queens
28	30.0	0.0	4	0.0	Broad Channel	Queens
25	27.0	0.0	6	0.0	Breezy Point/Fort Tilden/Riis Beach	Queens
57	58.0	0.0	10	0.0	Country Club	Bronx
58	59.0	0.0	1	0.0	Crotona Park	Bronx
97	99.0	0.0	1	0.0	Freshkills Park	Staten Island
105	111.0	0.0	3	0.0	Green-Wood Cemetery	Brooklyn
104	109.0	0.0	3	0.0	Great Kills	Staten Island
196	206.0	0.0	4	0.0	Saint George/New Brighton	Staten Island
211	221.0	0.0	4	0.0	Stapleton	Staten Island

- High ratio zones often serve as departure points for longer trips or areas with many residents and workers commuting, such as airports, residential neighborhoods, and villages.
- low ratio zones are typically popular destinations, including entertainment districts like parks, beaches, and clubs or large commercial areas where passengers are dropped off.

3.2.7. Identify the top zones with high traffic during night hours

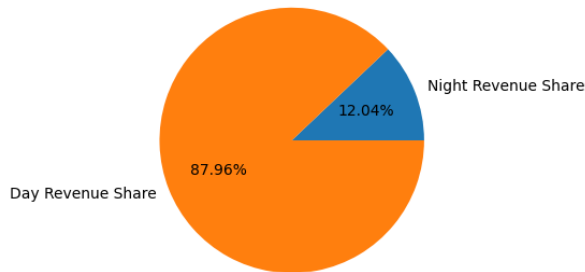
Top Pickup	PULocationID	zone	trip_count
50	79	East Village	2486
82	132	JFK Airport	2297
164	249	West Village	2008
28	48	Clinton East	1633
96	148	Lower East Side	1547

Top Dropoff	DOLocationID	zone	trip_count
1235	79	East Village	145
1689	107	East Village	134
3146	170	East Village	130
1253	79	Lower East Side	126
1278	79	West Village	119

During night hours, high pickup and dropoff activity is concentrated in entertainment and residential areas like East Village, West Village, Lower East Side, and Midtown Center, indicating significant night-time economy activity

3.2.8. Find the revenue share for nighttime and daytime hours

Revenue Share: Nighttime vs Daytime



- **Night hours** (11 PM - 5 AM) contribute approximately **12.04%** to the total revenue, while **day hours** (6 AM - 10 PM) account for a substantial **87.96%**.
- This highlights that while night hours are important, the bulk of the revenue comes from day-time operations.

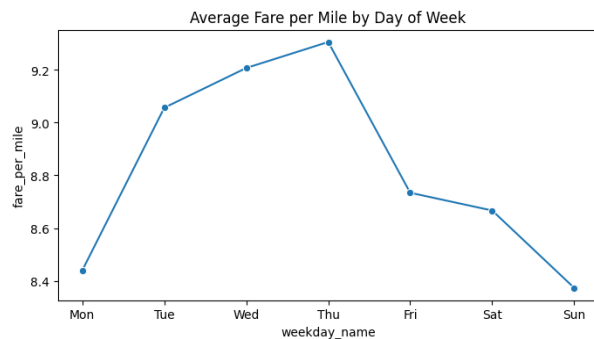
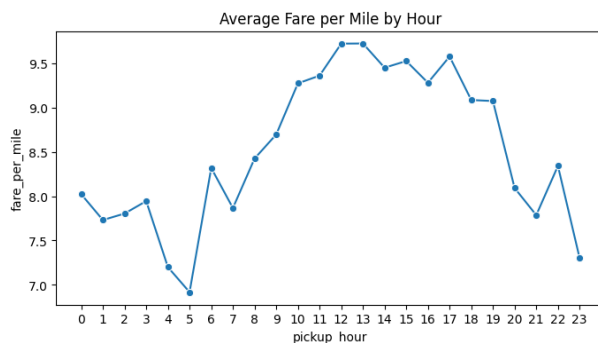
3.2.9. For the different passenger counts, find the average fare per mile per passenger

passenger_count	fare_per_mile	fare_per_mile_per_passenger
1.0	8.770658	8.770658
2.0	9.110312	4.555156
3.0	9.247927	3.082642
4.0	10.257903	2.564476
5.0	8.397846	1.679569
6.0	8.136741	1.356123

This metric shows relatively consistent values across different passenger counts, suggesting a base fare structure that isn't heavily adjusted per additional passenger. For example, 1 passenger: \$8.77/mile/passenger, 2 passengers: \$4.55/mile/passenger, 6 passengers: \$1.35/mile/passenger

$$\text{fare_per_mile_per_passenger} = \text{fare per mile} / \text{passenger count}$$

3.2.10. Find the average fare per mile by hours of the day and by days of the week

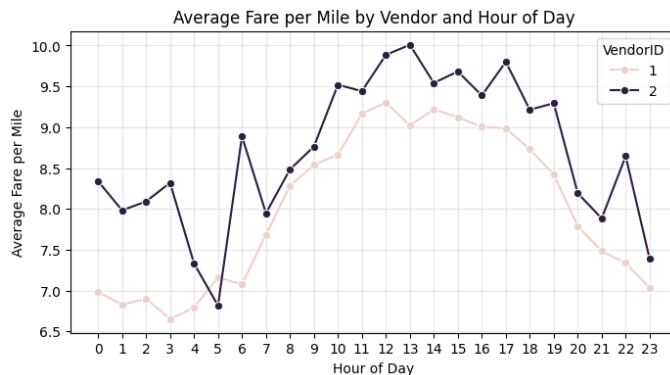


Average Fare per Mile by Hour:

- Fare per mile is lowest during early morning hours (4–6 AM).
- Prices steadily rise through the morning and peak between 10 AM and 7 PM, showing the highest fare-per-mile rates.
- Evening hours (6–10 PM) show a moderate decline, followed by another dip late at night.
- Overall, **midday trips** are the most expensive per mile, while early mornings are the cheapest.

Average Fare per Mile by Day of Week:

- Fare per mile is higher on weekdays, especially **Tuesday to Thursday**, indicating increased demand or slower traffic.
- Fares drop on Friday and continue declining through the weekend, with Sunday being the cheapest day.
- This suggests weekday business travel drives higher per-mile pricing, whereas weekend trips are more economical.

3.2.11. Analyse the average fare per mile for the different vendors

Vendor 2 generally exhibits a slightly higher average fare per mile compared to Vendor 1 across most hours of the day. This could indicate different pricing strategies or service levels.

3.2.12. Compare the fare rates of different vendors in a distance-tiered fashion

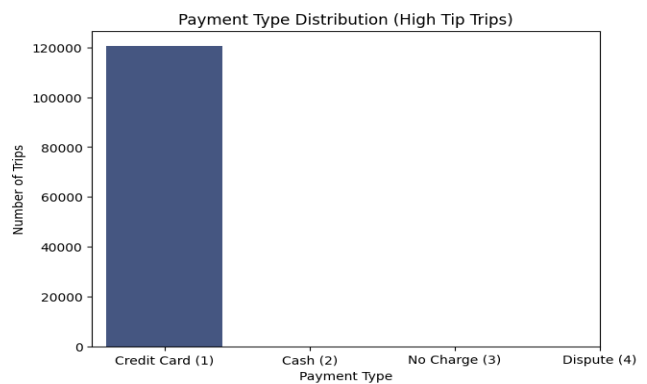
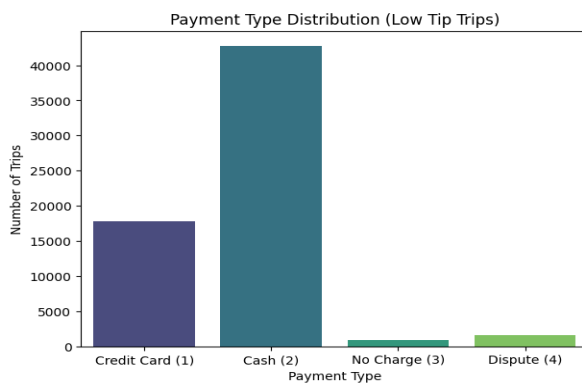
VendorID	distance tier	fare per mile
1	0-2 miles	9.605092
2	0-2 miles	10.547878
1	2-5 miles	6.342421
2	2-5 miles	6.507295
1	>5 miles	4.946539
2	>5 miles	5.059361

- 0-2 miles:** Highest fare per mile (e.g., for Vendor 1, around \$9.6/mile, for Vendor 2, around \$10.5/mile). This suggests a higher base fare component dominating shorter trips.
- 2-5 miles:** Lower fare per mile than short trips (e.g., for Vendor 1, around \$6.3/mile, for Vendor 2, around \$6.5/mile). The base fare impact is diluted by distance.
- >5 miles:** Lowest fare per mile (e.g., for Vendor 1, around \$4.9/mile, for Vendor 2, around \$5.5/mile). This indicates a more distance-driven pricing for longer journeys.

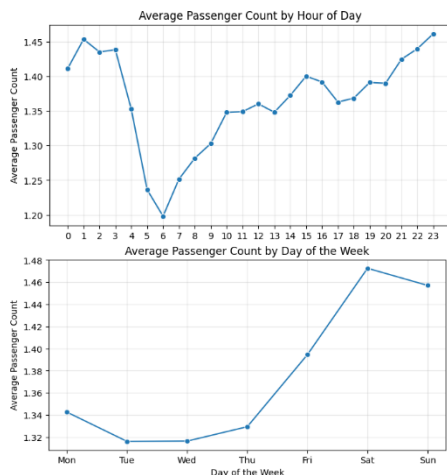
3.2.13. Analyse the tip percentages

distance_group	tip_percent
0–1 mile	24.398920
1–3 miles	20.617915
3–7 miles	17.486204

- the trip distance increases, the average tip percentage generally decreases. This could indicate that passengers are less inclined to tip a higher percentage for longer trips, possibly due to the higher total cost of the trip
- Passenger count doesn't show a strong, consistent correlation with tip percentage.
- This closely align with hourly trip count trend. Tip percentages are generally lower during early morning hours (e.g., 4 AM - 5 AM) and higher during late afternoon/evening hours. The most significant factor leading to low tip percentages is often the payment type (cash trips don't record tips).
- Low tip trips (e.g., <10% tip) are often associated with **cash payments**, potentially longer distances (where fare amount is higher, making a fixed tip percentage seem lower), and can occur at any hour. High tip trips (e.g., >25% tip) are almost exclusively linked to **credit card payments**, often involve shorter distances, and are more common during peak demand hours.



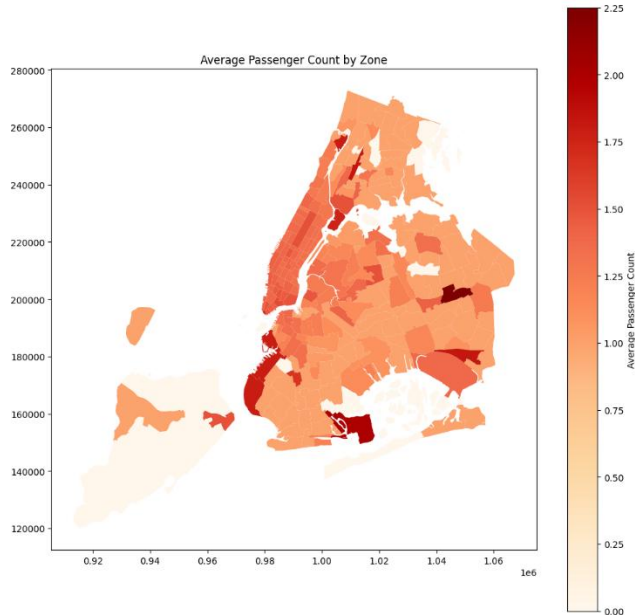
3.2.14. Analyse the trends in passenger count



Average passenger counts tend to be higher during **evening hours** and lower during **morning hours**,

Weekends, particularly Saturday, show a slightly higher average passenger count than weekdays, suggesting group travel for social activities on weekends

3.2.15. Analyse the variation of passenger counts across zones



Zones like Newark Airport (EWR) and Willets Point exhibit higher average passenger counts, indicating these locations are often used for group travel (e.g., airport transfers, event venues). Many Bronx zones show lower average passenger counts, closer to 1.

3.2.16. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.

- Prevalence of Surcharges:
 - extra: 154258 trips (53.28%)
 - congestion_surcharge: 241236 trips (83.32%)
 - airport_fee: 2572 trips (0.89%)
- Top pickup zones with surcharges:

	PULocationID	surcharge_count	LocationID	zone
83	132	14862	132.0	JFK Airport
155	237	13762	237.0	Upper East Side South
103	161	13555	161.0	Midtown Center
154	236	12292	236.0	Upper East Side North
104	162	10534	162.0	Midtown East

- Top dropoff zones with surcharges:

	DOLocationID	surcharge_count	LocationID	zone
226	236	12974	236.0	Upper East Side North
227	237	12272	237.0	Upper East Side South
154	161	11401	161.0	Midtown Center
220	230	8819	230.0	Times Sq/Theatre District
163	170	8658	170.0	Murray Hill

- The frequency of surcharges by `pickup\_hour` showed that surcharges are more prevalent during standard business hours and evening peaks, coinciding with periods of higher traffic and demand when `extra` charges (e.g., rush hour) and `congestion\_surcharge` are most applicable.

4. Conclusions

4.1. Final Insights and Recommendations

4.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

- Strategically position more taxis in high-demand areas (e.g., Midtown, Upper East Side, JFK Airport) during peak hours (17:00-19:00 on weekdays, later evenings on weekends)([section 3.2.2](#) and [3.2.3](#)).
- For identified "slow routes" (e.g., Route 234-256 at 18:00), integrate real-time traffic data to recommend alternative routes to drivers. Consider pre-emptive traffic management to minimise travel time and improve service efficiency. ([section 3.2.1](#))
- Implement surge pricing in high-demand zones during daytime peak periods. ([section 3.2.5](#))
- Offer incentives to drivers for operating in historically slow areas during off-peak hours to improve service coverage and reduce wait times.
- Repositioning algorithms can be implemented for cab positioning to address demand surges in specific areas, taking into account current traffic, historical speeds, and predicted demand to minimise trip duration and fuel consumption.

4.1.2. Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months.

- During weekday morning (7-10 AM) and evening (4-7 PM) peaks, concentrate cabs in major business districts and residential areas with high outbound commuter traffic (e.g., Midtown, Financial District, Upper East Side/West Side).
- Shift cab distribution towards entertainment hubs, tourist attractions, and residential areas with high evening social activity on weekends (e.g., East Village, West Village, Times Square).
- Maintain a consistent and adequate supply of taxis at JFK and LaGuardia Airports, especially during peak travel times. Utilize the high passenger counts at these locations to ensure efficient allocation.
- Deploy more empty cabs to high pickup ratio zones (origins) and encourage drivers to move from high dropoff ratio zones (destinations) to nearby high-demand pickup areas.([section 3.2.6](#))
- For night hours (11 PM - 5 AM), position cabs strategically in popular nightlife areas and residential zones with high night-time activity like East Village, West Village, and parts of

Midtown. Implement surge pricing (night charges) during these hours to incentivize drivers.([section 3.2.7](#))

4.1.3. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.

- July–September show low revenue; offering competitive or promotional pricing during these months and off-peak hours can boost pickups and overall earnings.([section 3.1.4](#))
- Correlation between trip duration and fare is low (0.29). The company can impose a waiting charge for the ride, which will increase the correlation between these two variables.([section 3.1.7](#))
- If Vendor 2 consistently charges higher fares per mile, and customers are still choosing them, it might indicate a perceived higher service quality. Investigate if there are opportunities to match or slightly undercut Vendor 2's pricing while improving service. If Vendor 1 has a lower fare per mile, market this as a competitive advantage.([section 3.2.11](#))
- Re-evaluate the fare-per-mile structure for different distance tiers. While shorter trips (0-2 miles) currently have a higher fare/mile, consider optimizing the pricing for longer trips (>5 miles) to ensure competitiveness, especially if Vendor 2's pricing is more favorable in these segments.([section 3.2.12](#))
- Implement programs to encourage higher tipping behavior, especially in scenarios associated with low tips (e.g., cash payments, early morning rides). This could involve driver quality incentives or promotional messaging to customers.([section 3.2.13](#))